

IMO(54th) – 2013

First Day

1. Prove that for any pair of positive integers k and n there exist k positive integers $m_1, m_2, \dots, m_k$ (not necessarily different) such that $1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right) \left(1 + \frac{1}{m_2}\right) \dots \left(1 + \frac{1}{m_k}\right)$.
2. A configuration of 4027 points in the plane is called Colombian if it consists of 2013 red points and 2014 blue points, and no three of the points of the configuration are collinear. By drawing some lines, the plane is divided into several regions. An arrangement of lines is good for a Colombian configuration if the following two conditions are satisfied:
 - (a) No line passes through any point of the configuration;
 - (b) No region contains points of both colors.
 Find the least value of k such that for any Colombian configuration 4027 points, there is a good arrangement of k lines.
3. Let the ex-circle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1 . Define the points B_1 on CA and C_1 on AB analogously, using the ex-circles opposite B and C , respectively. Suppose that the circumcentre of the triangle $A_1B_1C_1$ lies on the circumcircle of the triangle ABC . Prove that triangle ABC is right angled.

Second Day

4. Determine all ordered pairs (m, n) of positive integers such that $\frac{n^3 + 1}{mn - 1}$ is an integer.
5. Let S be the set of real number strictly greater than -1 . Find all functions $f : S \rightarrow S$ satisfying the two conditions:
 - (a) $f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all x and y in S ;
 - (b) $\frac{f(x)}{x}$ is strictly increasing on each of the intervals $-1 < x < 0$ and $0 < x$.
6. Show that there exist a set A of positive integers with the following property: For any infinite set S of primes there exist two positive integers $m \in A$ and $n \notin A$ each of which is a product of k distinct elements of S for some $k \geq 2$.

**